

2026 Innovative Executives

Learn how top leaders
are building future-ready
enterprises



Red Hat
Innovation Awards



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Foreword: from Brian Gracely, Senior Director for Portfolio Strategy, Red Hat

Making the vision a reality: it's time to take inspiration from innovation

After talking with IT leaders for the better part of 2 decades—through the rise of cloud, the container revolution, and now the AI wave—I have noticed something consistent: technology is rarely what stops people. What stops them is the fear of breaking something that is already working.

That fear is legitimate.

Every enterprise in this book is running real operations. Banking transactions that cannot fail. Military health systems where downtime has consequences. Manufacturing lines where a bad deployment cascades into physical disruption. These are not greenfield startups with the luxury of burning it down and rebuilding. They are organizations with real stakes—and they modernize anyway.

What I find most interesting about these stories is not the technology choices themselves, though those matter. It is the approach. None of these teams threw everything out and started over. They modernized progressively, at their own pace, without operational disruption. That pattern is something I have come to think of as the real competitive advantage. Not the AI. Not the containers. The discipline to keep moving forward without breaking what the business is dependent on today.

This year's list of Innovative Executives each found their own path through that challenge. Their stories are well worth reading in full—but here is a sample of what they have accomplished:

[**ARSAT**](#) is providing more responsive services and increasing data sovereignty by augmenting network operations with a consistent AI platform.

[**The Defense Health Agency \(DHA\)**](#) is saving US\$21 million and reclaiming thousands of hours for higher-value strategic priorities and global humanitarian missions through enterprise-wide automation.

[**Emirates NBD**](#) has consolidated 9,000 virtual machines (VMs) and modern container workloads onto a single, AI-ready platform to ensure 24x7 banking resilience.

[**StarHub**](#) is building digital trust and delivering real-time quality of experience through cutting-edge polycloud architecture and agentic AI.

This is harder than it sounds. Hybrid environments are genuinely complex. The gap between traditional infrastructure and cloud-native architecture is real and managing both simultaneously while trying to move forward creates friction that can stall even well-funded teams. The organizations in this e-book found a way through that friction—and in most cases, the common thread is a consistent foundation that lets them move across environments without re-platforming from scratch every time something changes.

If you are reading this, you are probably somewhere in the middle of your own version of this journey. Maybe you are managing technical debt that is slowing down your AI ambitions. Maybe you have a hybrid environment that's increasingly hard to operate at scale. Maybe you're trying to figure out how to give your team back time that is currently going into keeping the lights on. The question worth sitting with is: which of these stories looks most like where you are trying to go?

The answers are not all the same—nor should they be. But the underlying principle usually is. Build a foundation consistent enough that innovation becomes the default, not the exception.

That is what these teams did. I hope it is useful to see how.

ARSAT provides more responsive services and better connectivity with Red Hat OpenShift AI

Implementing a consistent AI platform augments network operations, automates processes, and increases data sovereignty

ARSAT, an Argentine telecommunications infrastructure provider with more than 1,500 clients, needed to respond rapidly to changes in technology and unify its AI development. By deploying Red Hat® OpenShift® AI, the state-owned organization augmented the capabilities of its Network Operations Center (NOC) with AI. This deployment has allowed ARSAT to maximize efficiency, respond faster to client needs, and support data sovereignty while increasing network intelligence and expanding connectivity.

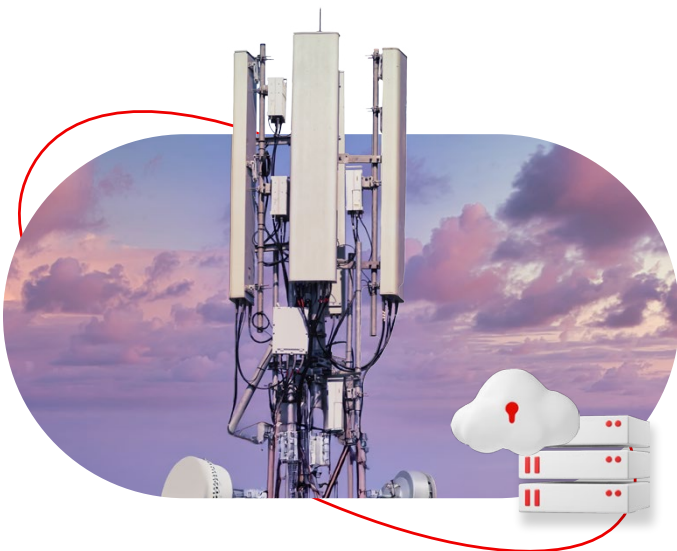
Adapting to technological change across disparate environments

ARSAT's core mission is to provide and expand high-quality connectivity and digital infrastructure across Argentina. However, the company's divisions, including its fiber networks, satellites, and datacenters, had grown autonomously over several years and relied on very different technologies. Because these divisions operated independently with different objectives and operational speeds, integration was difficult.

To provide clients with services in a more centralized and efficient way, ARSAT needed to bring these departments together to create complementary, integrated solutions that combined connectivity, processing, storage, and backup services.

To address this operational complexity and support more intelligent service delivery, ARSAT worked with Red Hat to implement Red Hat OpenShift AI, selecting the solution for its open source nature.

Red Hat OpenShift AI now provides a consistent platform that centralizes AI data and standardizes operations across multiple business verticals. The platform actively augments NOC operators—who manage 1,200 sites and 35,000 km of fiber optics—by assisting with the triage, diagnosis, and resolution of operational issues. This supports faster and more effective responses while preserving human decision making.



Strengthening network intelligence and service innovation

Accelerated client identification and issue resolution

The implementation of AI has allowed ARSAT to resolve initial points of contact with clients much more rapidly. By identifying a client and searching for their specific history, ARSAT can better understand the unique architecture of each of its services. This allows the team to perform a more accurate triage of the failure and provide a swifter resolution to key issues.

Improved internal automation and operator experience

ARSAT has successfully incorporated the need to automate processes using AI assistance. For NOC operators, this technology aids in diagnostics, making workloads not only simpler and faster but also more efficient and enjoyable. AI is also used to augment operators' intelligence, helping them respond in closer to real time while preserving human decision making.

Strengthened technological sovereignty and new skill generation

The project has had a significant positive effect on ARSAT's technological sovereignty. By developing AI capabilities in house, ARSAT is generating new skills within its teams to manage data connectivity, which is the main component of its operations. This internal capacity building enables ARSAT to offer its clients more effective digitization strategies and local knowledge exploitation.

Expanding AI to new products and infrastructure services

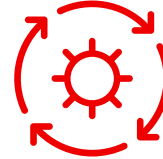
Having successfully incorporated AI capabilities within its NOC, ARSAT next plans to use Red Hat OpenShift AI to support the development and launch of new products centered around its federal fiber optic network. The organization is now focused on developing future capabilities to further increase network intelligence, expand digital infrastructure services, and support innovation across Argentina's connectivity ecosystem.



Key outcomes



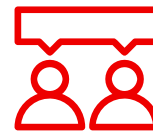
Supported NOCs in managing 1,200 sites and 35,000 km of fiber optic cable



Allowed more accurate triage of failure and swifter resolution to key issues



Reduced manual workloads while increasing employee efficiency and satisfaction



Enabled faster, more personalized responses to client needs

Red Hat products used

- Red Hat OpenShift AI





How AI is leading ARSAT from fragmented operations to augmented connectivity

by Mariano Greco, President and General Manager, ARSAT

ARSAT's mission is to provide and expand high-quality infrastructure and connectivity throughout Argentina and into space. We manage a massive scale of operations, including 36,000 km of fiber optics, 2 geostationary satellites covering the Americas, and a 4,500 m² tier 3 datacenter.

However, as we sought to respond to increasingly rapid technological changes, we faced a significant hurdle: our business verticals had grown very autonomously. Our satellite, network, and datacenter businesses used fundamentally different technologies, so providing centralized and complex services to our 1,500 clients was difficult.

To overcome these isolated operations, we needed core technology that could centralize our processes and allow us to view each client as a single entity. And because we have a strong internal open source culture, we naturally aligned with Red Hat and implemented Red Hat OpenShift AI.

We deployed this platform not just as an out-of-the-box product, but as a framework and centralizing engine to transform our Network Operation Centers (NOCs), which manage 1,200 sites nationwide. This implementation allowed us to incorporate AI to organize data and centralize operations, helping us understand each client's unique, complex architecture, much like medical history.

Since we brought the project into production in 2025, the results of this technological integration have been profound. By using AI as 'augmented intelligence', we are not replacing human decision making; rather, we are assisting our NOC operators with complex pattern recognition and preliminary diagnostics.

This support makes their workload simpler, faster, and more enjoyable. And for our B2B clients, this directly translates to a much faster resolution at the initial point of contact and highly personalized triage for their specific operational failures.

Ultimately, this transformation is about more than just operational speed; it has a welcome effect on our technological sovereignty. By developing these AI capabilities in house, we are generating crucial new skills within our teams to manage data connectivity, empowering us to offer clients more effective, localized digitization strategies.

As we look to the future, we are preparing to expand this AI engine to our Federal Fiber Optic Network and associate it with new commercial products directly at the point of contact. More importantly, maintaining our flexibility and working as a community, we are ensuring that our technological growth continues to serve as an engine of humanity.



DHA saves US\$21m and moves from manual firefighting to strategic priorities

Replacing fragmented systems with Red Hat automations ensures financial benefits and reclaims thousands of hours for higher-value work

Facing budget reductions and manual inefficiencies, the US Defense Health Agency (DHA) collaborated with Red Hat to automate its infrastructure using Red Hat Ansible® Automation Platform. This transformation saved US\$21 million and reduced 1-hour tasks to 5 minutes; as a result, engineers now focus on development rather than maintenance. By standardizing operations, DHA converted manual medical records to digital and improved care for 1.3 million service members. The project also supports global humanitarian missions aboard USNS hospital ships, enhancing operational readiness and patient outcomes.

Standardizing infrastructure to ensure sustainability

To address manual inefficiencies and fragmented systems, the DHA worked with Red Hat to standardize its infrastructure through enterprise-wide automation. The core solution involved deploying Red Hat Ansible Automation Platform to replace disparate custom-built solutions with a standardized, automated datacenter environment.

Led by Process Automation Lead Dave King, the team implemented Infrastructure as Code (IaC) to develop a unified automation framework.

This implementation focused on democratizing automation, utilizing Ansible Automation Platform's accessible YAML syntax to bridge the technical gap between seasoned sysadmins and new DevOps engineers, and Red Hat Customer Portal Labs and Ansible Playbooks to access practical, hands-on training. To manage the agency's heavy regulations, the team adopted a Policy as Code approach, wrapping legacy infrastructure in modern management layers to automatically enforce Security Technical Implementation Guide (STIG) compliance and Risk Management Framework (RMF) standards.

To ensure long-term sustainability, the DHA mandated standardization through open collaboration, requiring that all automation assets be modular, reusable, and stored in version-controlled repositories. The agency is now able to deploy compliance updates and changes simultaneously across the entire Military Health System as automated background tasks.



Improving efficiency and enhancing deployment

Increased efficiency and promoted proactive development

The implementation reduced the time required for maintenance tasks—which previously took an hour of manual work—down to just 5 minutes. This increased efficiency allowed the DHA to shift from reactive firefighting to proactive work. By automating repetitive tasks, the agency also freed its engineers to focus on higher-value mission priorities rather than routine maintenance.

Reduced costs over 5 years

The project resulted in a US\$21 million financial saving by selecting Red Hat automation solutions over previous traditional VMware products. The total projected saving over the next 5 years is US\$62 million.

Achieved faster, more coordinated, reliable deployments

Automation has enabled the conversion of manual medical records to a digital format, eliminating the need to physically transport files with soldiers, and ensuring that data is accessible globally. This reliability extends to humanitarian missions; the standardized infrastructure now supports the USNS Comfort and Mercy hospital ships, facilitating life-changing surgeries for underserved populations worldwide.



“We recovered our most valuable resource: time. We have reclaimed thousands of man-hours, allowing our 100+ engineers to pivot from repetitive toil to high-value innovation.”

Dave King

Process Automation Lead, Defense Health Agency

Expanding automation to new disciplines and projects

DHA now plans to expand Ansible Automation Platform to additional technical disciplines and new AI projects. Future initiatives include replacing SUSE Rancher, completing a full US\$62 million VMware takeout, and extending automation to the USNS Mercy and Comfort hospital ships to further support global humanitarian missions.

Timeline

• **Implementation officially initiated:**

March 2025

• **Environment standardization work:**

March–July 2025

• **Production started:**

July 2025

• **AI project discussions and launch:**

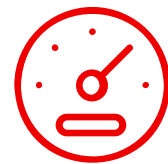
Late 2025



**US\$21 million in immediate
cost avoidance**



**US\$62 million total
projected 5-year saving**



**92% reduction
in task time**

Red Hat products used

- Red Hat Ansible Automation Platform
- Red Hat Training
- Red Hat Learning Subscription





How automation is leading the DHA from firefighting to innovation

by Dave King, Process Automation Lead, Defense Health Agency

At the Defense Health Agency, our mission is absolute: we manage healthcare for 1.3 million active-duty service members across the Army, Navy, Air Force, Marines, and Coast Guard. However, in an environment defined by budget reductions and workforce constraints, we recently faced a critical tipping point. Our engineers were burdened by manual, error-prone infrastructure tasks, leaving us constantly fighting fires rather than proactively implementing strategy.

To maintain operational readiness, we couldn't just work harder; we had to change how we worked. That's why we collaborated with Red Hat to standardize our datacenters through enterprise-wide automation.

My focus as an architect was to democratize automation. We utilized Red Hat Ansible Automation Platform to implement Infrastructure as Code (IaC), using accessible syntax to bridge the gap between seasoned sysadmins and new DevOps engineers. By wrapping our legacy infrastructure in modern management layers, we adopted a Policy-as-Code approach. This allows us to enforce strict security standards (STIG) and Risk Management Framework (RMF) compliance automatically, ensuring our systems remain audit-ready without manual intervention.

The results of this cultural and technical shift have been transformative. By choosing Red Hat over legacy VMware solutions, we achieved US\$21 million in immediate cost avoidance, with a projected US\$62 million in savings over the next 5 years.

More importantly, we recovered our most valuable resource: time. Maintenance tasks that previously required 1 hour of manual work now take just 5 minutes.

We have reclaimed thousands of man-hours, allowing our 100+ engineers to pivot from repetitive toil to high-value innovation.

Ultimately, value in our environment is defined by service availability for the medic, the physician, and the patient. Automation has enabled us to digitize medical records effectively, eliminating the need for soldiers to carry physical folders and making critical data accessible globally. We are now extending these capabilities to the USNS Comfort and Mercy hospital ships, ensuring reliable IT infrastructure supports humanitarian surgeries for underserved populations worldwide.

We have shifted from reacting to mandates to fostering a mindset of continuous evolution. That's why today, our infrastructure is no longer a bottleneck; it is a dependable partner in patient care.



Emirates NBD boosts efficiency by unifying VMs, containers, and AI with Red Hat

Consolidating 9,000 VMs and modern container workloads onto a single platform provides enterprise-grade resilience, operational efficiency, and rapid innovation

Emirates NBD, a major banking group serving more than 20 million customers, faced operational complexity from running parallel infrastructures for traditional core systems and modern applications. To support 24x7 banking demands and an expanding market, the bank collaborated with Red Hat to consolidate its VMs and container workloads using Red Hat OpenShift Virtualization. This transformation eliminated operational separation, enabled the migration of up to 140 machines per night, and provided a scalable foundation ready for AI.

Managing complexity across critical banking infrastructure

Every day, millions of transactions flow through Emirates NBD's infrastructure, from retail payments to real-time credit decisions. Because mobile platforms, APIs, and core systems must operate 24x7, downtime severely affects customers, business, and the wider economy.

Historically, the bank operated multiple parallel infrastructures: a traditional virtualization stack for core systems and a separate container platform for modern applications. This approach led to fragmentation, creating disparate operations, duplicated tooling, inconsistent controls, and higher costs.

Following the acquisition of a major bank in India, Emirates NBD recognized that standardizing on a single, centralized platform would be easier to manage, better for customers, and support greater scalability.

Grounded in open standards and a desire for long-term architectural sovereignty, Emirates NBD adopted Red Hat OpenShift Virtualization to manage both VMs and containers on a single platform. Working alongside Red Hat Consulting, the bank took a structured approach to migrating workloads, using the Red Hat's migration toolkit for virtualization to minimize risk and potential customer downtime.

Starting with less-critical workloads before moving to core banking systems, the team scaled processes to migrate up to 140 machines per night, eventually operating around 9,000 VMs on the unified platform. This architectural convergence reduced complexity, standardized observability and security, and created a consistent developer experience across the bank's 3,000 engineers.



Delivering greater scalability, efficiency, and innovation

Increased operational efficiency and cost predictability

By consolidating VMs and containers, Emirates NBD achieved unified orchestration, integrated networking and storage, and standardized continuous integration and continuous delivery (CI/CD) pipelines. Centralized visibility has allowed the bank to better measure use and capacity, supporting more informed infrastructure management, while using Red Hat as a single vendor has increased cost transparency and predictability.

Achieved faster, reliable product deployments

Red Hat OpenShift enables Emirates NBD to release new products to the market more quickly—ranging from cryptocurrency offerings to wealth management assets, such as shares and bonds—while maintaining strict 24x7 availability.

Removed the false choice between legacy and modern infrastructure

By running traditional business-critical VMs side by side with AI-native container workloads on the same Kubernetes platform, the bank achieved strategic optionality without disruption. This allows Emirates NBD to modernize progressively at its own pace while ensuring production stability and lowering operational complexity.

Expanding automation to AI and new initiatives

Emirates NBD is expanding its capabilities by exploring Red Hat OpenShift AI to maximize the value of its graphics processing units (GPUs). This open source approach gives the bank the flexibility to use preferred GPUs and run AI workloads wherever necessary—on premise or in the cloud—creating a foundation that is ready for next-generation banking innovation.



Key outcomes



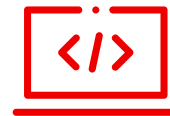
**More than 20 million
customers served globally**



**Around 9,000 VMs operated
on the new unified platform**



**Achieved a migration velocity
of up to 140 machines per night**



**Delivers consistent developer experience
for 3,000 internal engineers**

Red Hat products used

- Red Hat Enterprise Linux®
- Red Hat OpenShift
- Red Hat OpenShift Virtualization
- Red Hat OpenShift AI
- Red Hat Consulting
- Red Hat Technical Account Manager





From fragmentation to AI-ready innovation: Emirates NBD achieves architectural convergence

by Miguel Rio-Tinto, Chief Digital and Information Officer, Emirates NBD

Emirates NBD operates on a global scale, serving more than 20 million customers across multiple geographies. Every single day, millions of transactions flow through our infrastructure—from retail payments and corporate trade finance to real-time credit decisions and digital onboarding journeys. Our mobile platforms, APIs, and core systems operate 24x7, so downtime isn't just a technical inconvenience: it directly affects our customers, businesses, and the wider economy.

We faced a challenge that is common to many large banks: operating multiple parallel infrastructures. With traditional virtualization stacks for core systems and separate container platforms for modern applications, we were dealing with fragmentation, duplicated tooling, inconsistent security controls, and a very high cost to serve.

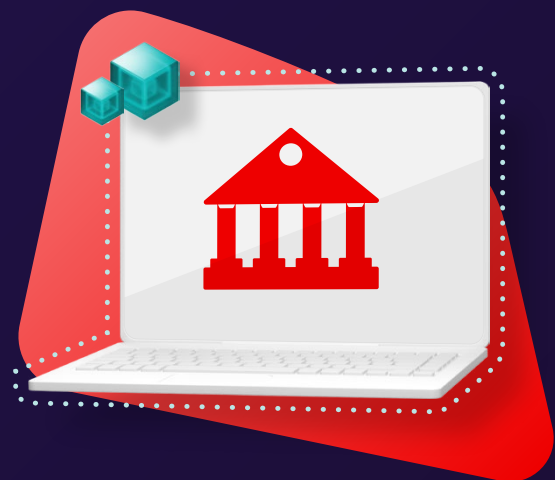
To maintain the enterprise-grade resilience required at our scale, we couldn't just continue to manage disparate operations: we had to pursue architectural convergence. That is why we partnered with Red Hat to create a single, unified platform capable of running both VMs and containers.

Our modernization journey is built on open standards, portability, and long-term architectural sovereignty—we want to control our own engineering destiny. By adopting Red Hat OpenShift Virtualization, we now run our 9,000 traditional VMs and cloud-native containers side by side on the same Kubernetes control plane. This shift has allowed us to implement unified orchestration, consistent policy enforcement, integrated networking and storage, and standard CI/CD pipelines.

The results of this shift have been transformational: instead of maintaining 2 operating models, we now have just 1. This has dramatically reduced complexity, standardized our observability, and created a consistent developer experience across our 3,000 engineers. We have also been able to modernize progressively, migrating critical banking workloads at our own pace while maintaining essential production stability.

Ultimately, in banking, stability and innovation must coexist. Probably the most powerful benefit of OpenShift Virtualization is that it removes the false choice between traditional and modern infrastructure. We can now run mission-critical VMs and AI-native container workloads on the exact same platform. This means faster innovation, lower complexity, and a foundation that is fully ready for AI at scale.

And by shifting toward this programmable, self-service infrastructure, we are also giving our engineers the freedom to focus on higher-order problem solving, ensuring that transformation becomes continuous rather than episodic.



One New Zealand accelerates innovation and cuts deployment times by 40%

Modernizing a fragmented IT infrastructure with automated, cloud-native platforms delivered by Red Hat supports 5G rollout, resulting in major cost savings

Facing a complex legacy environment built through multiple acquisitions, telecommunications service provider One New Zealand needed a scalable foundation to support the next 5 to 10 years of connectivity services. By collaborating with Red Hat to establish the One Horizontal Cloud platform, powered by Red Hat OpenShift and Red Hat Ansible Automation Platform, the company unified its fragmented architecture. This transformation reduced delivery times by 40% and operating costs by up to 45%, empowering engineers to focus on value-added services and enabling the deployment of a fully cloud-native 5G core.

Building a scalable, future-ready foundation

In a market that demands uninterrupted connectivity, One New Zealand's highly virtualized but fragmented infrastructure hindered its ability to innovate and scale. Voice, messaging, and 5G core platforms operated on separate architectural frameworks, creating isolated structures and engineering challenges.

To resolve this, One New Zealand worked with Red Hat Consulting to design a standard reference architecture and build the One Horizontal Cloud.

At the core of this unified backbone is Red Hat OpenShift, supported by Red Hat Ansible Automation Platform and Red Hat Advanced Cluster Management for Kubernetes, which provide end-to-end automation and reliable service delivery.

One New Zealand has also invested in its people, using Red Hat Learning Subscription and classroom-based training with a gamified approach to help engineers transition to container-based architectures.



Improving provisioning and enhancing deployment

Increased efficiency with zero-touch processes

By implementing uninterrupted automation and zero-touch provisioning, One New Zealand can automatically provision everything from bare-metal servers to operating systems, cluster loading, and security configurations. This end-to-end lifecycle automation has shifted engineers away from manual configuration tasks toward proactive, value-added work for the business. The unified platform has also established a common language, breaking down barriers between networking, application, and cloud infrastructure teams.

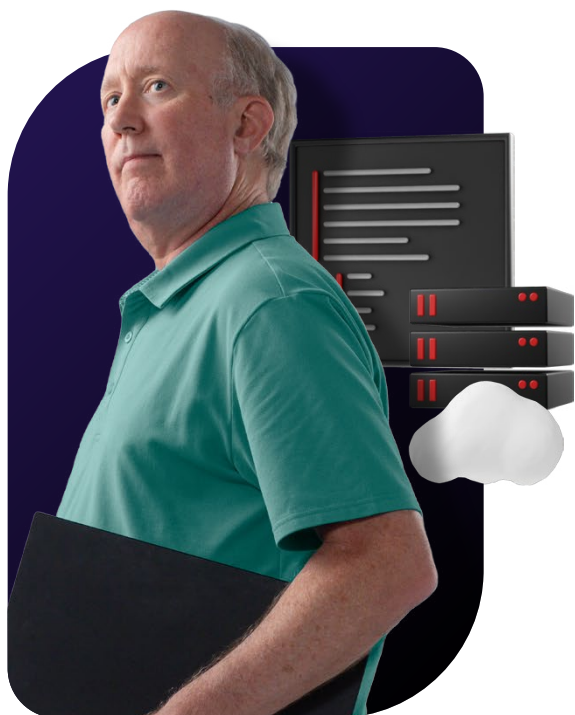
Reduced costs and accelerated service delivery

The standardized platform has resulted in a 40% reduction in delivery times and a 30-45% decrease in operational costs for deploying new workloads. Projects that previously took weeks or months to deliver can now be completed in days.

In-service software upgrades are now 40 times faster than they were on the legacy platforms, providing a highly standardized and reliable upgrade process.

Achieved a unified architecture for business-critical networks

One Horizontal Cloud serves as the cohesive foundation for One New Zealand's most critical applications, including its voice and messaging platforms, subscriber databases, and a new 5G standalone core. Transitioning these key elements of the telco stack to a single cloud-native mobile core has allowed the company to move at market speeds and continuously deliver new, highly resilient digital products to 2.3 million mobile connections and more than 110,000 business, enterprise, and government customers.



Expanding automation to new disciplines and AI innovation

One New Zealand now plans to use its automated infrastructure to become the world's most AI-enabled telecommunications operator. The organization is expanding the use of the platform to deploy artificial intelligence use cases across its network and IT environments, aiming to augment human capabilities in software development and network operations. One New Zealand is also exploring further collaborations with Red Hat surrounding digital sovereignty and autonomous network operations observability, viewing AI as a tool to amplify the capabilities of human engineers.

Key outcomes



**40% reduction in delivery times
for new workloads**



**30-45% decrease in operational
costs for deployments**



**40x faster in-service
software upgrades**



**2.3 million mobile connections and more
than 110,000 customers**

Red Hat products used

- Red Hat OpenShift
- Red Hat Ansible Automation Platform
- Red Hat Advanced Cluster Management for Kubernetes
- Red Hat Consulting
- Red Hat Learning Subscription





The agility to innovate: One New Zealand builds a future-ready network

by Kieran Byrne, Chief Technology Officer, One New Zealand

Telecommunications is an industry under constant change. The changing digital landscape dictates that we will no longer just be providing traditional mobile phone services. We are entering an era where millions—even billions—of devices need to be connected.

As Chief Technology Officer at One New Zealand, I oversee our network, IT infrastructure, cybersecurity, and wholesale business. Defining value in this digital-first environment is fairly simple: we need to win at market, and to do so, we need to be innovative, efficient, and resilient. We run business-critical infrastructure that the New Zealand economy relies on. Trust is, therefore, non-negotiable.

We need to prepare our business for the next 5 to 10 years of connectivity services. However, like many telecommunications service providers, our complex environment, built over multiple years through various acquisitions, was holding us back. This lack of cohesion made it incredibly difficult to innovate, achieve speed to market, and continuously deliver reliable services to our customers.

We needed a unified backbone. After an extensive market search, we chose to partner with Red Hat because it offered an open, scalable, telco-grade platform that allows us to integrate services from all our providers and containerize workloads in a standardized way.

The result is the One Horizontal Cloud, the foundation of our ambition to transition to a completely cloud-native network and IT infrastructure.

At the core of this transformation is Red Hat OpenShift, supported by Red Hat Ansible Automation Platform and Red Hat Advanced Cluster Management for Kubernetes. And through our partnership with Red Hat Consulting, we gained access to global best practices and reference architectures, allowing us to develop new operating models and truly maximize the platform's potential.

This transformation has been a game changer. By implementing end-to-end automation and zero-touch provisioning, we can now automatically provision everything from bare-metal servers to operating systems and configurations.

The One Horizontal Cloud is the foundation for our 5G services. We are currently deploying our 5G standalone core onto this platform—our first deployment of a full cloud-native mobile core—alongside our critical voice, messaging, and subscriber databases.

Innovation is deeply rooted in our DNA, but moving to container-based architecture requires a massive shift in mindset. That is why we invested heavily in our engineering teams using Red Hat Learning Subscription and classroom-based, gamified training. Our networking, application, and cloud infrastructure teams have broken down historical isolated structures and now collaborate using a common language and standard architecture.

We have set a bold goal: to be the most AI-based telco in the world. We are actively deploying AI use cases across our network and IT environments to augment human capabilities in software development and network operations. We also look forward to extending our partnership with Red Hat in emerging areas like digital sovereignty to ensure long-term resilience.



StarHub advances its cloud evolution toward a trusted digital infrastructure model

Singapore telco deploys strategy based on polycloud, agentic AI, and Red Hat OpenShift to build digital trust, enhance quality of experience, and strengthen enterprise performance

StarHub, a Singapore-based connectivity and digital services provider, is evolving. It is integrating polycloud agility and digital trust to transition from a traditional carrier into a trusted service provider grounded in resilient, scalable infrastructure. A key part of this process is the adoption of a polycloud architecture that intelligently distributes workloads across environments for optimal real-time quality of experience (QoE). The new foundation supports agentic AI, employing eSIM-like verification to monitor models and mitigate security risks, and blind computing to ensure data privacy.

Establishing a resilient framework for decentralized AI

StarHub is executing its polycloud and agentic AI strategy by transitioning from hardware-centric systems toward a cloud-native, software-defined architecture centered on Red Hat OpenShift. This platform is acting as a solid foundation for StarHub's transformation, providing an autonomous, access-neutral, hybrid environment.

The aim is to facilitate the adoption of agentic AI frameworks that can be supported across core to edge, with multiple cloud providers in between.

This will then allow StarHub to deliver QoE to its users in real time by selecting the most appropriate compute, hyperscaler, or edge platform, while also controlling costs.

Crucially, StarHub is using Red Hat OpenShift to implement blind computing through confidential containers. In this setup, Red Hat OpenShift provides the Secure Compute Boundary and Trusted Execution Environment (TEE), allowing encrypted data to be processed in isolated memory without being exposed to the host.

Beyond the technology layer, StarHub is actively collaborating with Red Hat's Telco Center of Excellence to develop AI security standards and automation, contributing these innovations back to the open source community to establish a resilient framework for decentralized AI operations.



Achieving a fundamental operational enterprise shift

Enhanced operational agility

StarHub has seen a reduction in time to market for new digital services. This is principally a result of its shift to agile, open source architecture.

Transformed StarHub's business model

StarHub has increased cost flexibility by adopting cloud-based operating models alongside traditional infrastructure investments. In doing so, it is also moving away from rigid, hardware-defined infrastructure to become a dynamic TechCo.

Delivered financial growth

StarHub has strengthened enterprise revenue performance, supported by its continued investment in scalable digital platforms. The result is underpinned by the strategic convergence of polycLOUD agility, innovative agentic AI, and cross-domain automation, which StarHub defines as the pillars of its success.



"At StarHub, innovation is not about speed alone. It is about building trust into the foundation. When resilient architecture and disciplined governance come first, we can scale AI sales, evolve predictably, and deliver lasting value. Trust is the infrastructure beneath every digital service that we build."

Dr. Volkan Sevindik

Chief Technology Officer, StarHub

Extending the roadmap for AI and automation

StarHub is now looking to continue its evolution, from TechCo into a trusted service provider, establishing institutional trust as the foundation for its next decade of growth. Future initiatives include integrating post-quantum cryptography and securing agentic AI to set the pace for sustainable innovation.

Timeline

- **Initiated:**
5-year roadmap
- **Adopted:**
PolycLOUD architecture
- **Integrated:**
Secure agentic AI
- **Continued:**
Evolution to becoming a trusted service provider



**Significant increase in
enterprise revenue**



**Reduction in time
to market**



**Significant volumes of
business-critical network
data are managed daily**

Red Hat products used

- Red Hat OpenShift
- Red Hat Ansible Automation Platform
- Red Hat Advanced Cluster Management for Kubernetes
- Red Hat Consulting
- Red Hat Technical Account Manager
- Red Hat Training
- Red Hat Learning Subscription





From connectivity to trusted digital infrastructure

by Dr. Volkan Sevindik, Chief Technology Officer, StarHub

The role of a telecommunications provider is being redefined. At StarHub, we recognize that remaining competitive in the next decade requires more than expanding connectivity: it requires strengthening trust at every layer of our digital infrastructure. Our evolution toward becoming a trusted service provider is not a branding exercise. It is a deliberate architectural shift built on 2 converging forces: polycloud agility and secure AI governance.

Our journey began by dismantling the idea that “cloud” simply means moving data to a public server. We have moved into the second phase of true polycloud architecture. Unlike standard hybrid models, our approach involves a horizontal cloud that intelligently selects the optimum compute environment for every specific workload in real time.

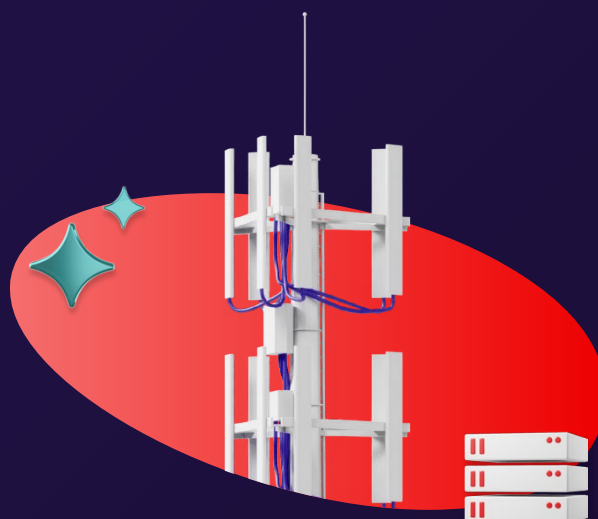
We are also no longer making decisions based solely on cost; we are optimizing for quality of experience. This architecture allows us to route low-latency applications to the edge while sending data-heavy analytics to the public cloud, creating the necessary readiness to support AI from the core to the edge. However, as we open our architecture, we must simultaneously lock down our systems. This is where we pioneer the concept of blind computing using Red Hat OpenShift.

In the age of agentic AI, data sovereignty is paramount. We are implementing architecture designed to minimize data exposure during storage, transfer, and processing through encryption and confidential computing techniques. By using confidential containers, we create a trusted execution environment.

This functions as a digital safe room where data is decrypted only within isolated memory for the exact moment of processing, ensuring that even the host machine never sees the raw information.

We are extending this institutional trust to the AI models themselves. We believe that an AI agent on our network should be vetted as rigorously as a human subscriber. That is why we are currently working to implement eSIM-like identity mechanisms for AI models, assigning them unique IDs and trust scores. This allows us to monitor model behavior in real time and take corrective action, including isolating connections should a model attempt to connect to a suspicious IP or exhibit rogue behavior.

For StarHub, innovation is not just about speed; it is about strengthening digital safeguards for government and enterprise clients. And by converging polycloud agility with these non-negotiable security guardrails, we are not just moving data—we are supporting Singapore’s digital resilience.



Tata Tele Business Services' journey to a cloud-native future

Replacing rigid traditional infrastructure with Red Hat OpenShift delivers financial growth and reduces time to market for new services

Tata Tele Business Services (TTBS) is a provider of telecommunications services and connectivity and cloud communication solutions for businesses in India. TTBS collaborated with Red Hat to modernize its Business Support System (BSS) and build its cloud-native communication platform, Smartflo. This transformation, built on Red Hat OpenShift Platform Plus and Red Hat OpenStack® Platform, allowed TTBS to optimize service roll-out times and gave teams the freedom to focus on digital-first innovation. As a result, TTBS reduced overall total cost of ownership (TCO) by around 25%, thereby contributing to positive earnings before interest, taxes, depreciation, and amortization (EBIDTA).

Modernizing traditional infrastructure to achieve business agility and scalability

To address fragmented, non-standardized architecture and reliance on a capital-intensive traditional hardware stack, TTBS sought a flexible, open standards-based cloud platform. It engaged with Red Hat to undertake the comprehensive BSS modernization initiative, moving it away from its existing cloud infrastructure.

The core solution involved deploying Red Hat OpenShift Platform Plus as the foundation on which to build Smartflo, TTBS's next-generation cloud communication platform. TTBS also worked very closely with its Red Hat Technical Account Manager to overhaul its technology landscape, helping the business through its transformation journey.

The new API-based solution now allows businesses to embed voice and communication capabilities directly into their applications without managing telephony infrastructure themselves. By adopting a layered, disaggregated architecture, TTBS eliminated vendor lock-in and optimized resource use across a common platform.



Building operational efficiency and promoting business growth

Accelerated service introduction and scaling

The transition to a cloud-native infrastructure allowed TTBS to respond quickly to evolving B2B communications demands from its customer base, leading to less customer churn, higher customer satisfaction, and an increase in value-added, tailored services.

Additionally, capacity expansion times were cut from days to hours, and overall cloud deployment time dropped from 4 weeks to only 1 week.

Reduced costs and optimized resources

By migrating to a shared cloud layer and replacing outdated hardware systems, TTBS achieved a TCO benefit of around 25%. By moving to a modular, hardware-, and vendor-agnostic architecture, TTBS was also able to optimize performance, avoid vendor lock-in, and unlock the full value of innovations at each layer.

Extreme automation implemented across build and run phases has streamlined operations and reduced manual intervention, delivering a public-cloud-like experience while lowering its carbon footprint and power requirements.

Increased revenue and enhanced customer experience

The resulting omnichannel Smartflo platform—which integrates Communication Platform as a Service (CPaaS), Contact Center as a Service (CCaaS), and Unified Communications as a Service (UCaaS)—has significantly enhanced customer experiences. Because of this strategic initiative and its resulting benefits, TTBS realized an increase in revenue generation, which contributed to positive EBITDA.



Extending open architecture to new services and AI capabilities

TTBS plans to use Red Hat OpenShift Platform Plus and Red Hat hybrid cloud design to continue its evolution. Future initiatives include building a DevOps framework with complete API integration to rapidly launch new services, such as API gateways and bare-metal provisioning. The flexible open architecture will also make it easier for TTBS to support advanced, AI-ready analytics and allow enterprise customers to run diverse, new-age workloads—including AI, machine learning, the Internet of Things, and big data—across various environments.

Timeline

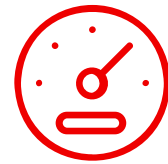
- **Implementation officially initiated:**
November 2024
- **Production started:**
April 2025



**25% reduction
in overall TCO**



**Service rollout time
reduced from months
to days**



**Capacity expansion
time reduced from
days to hours**

Red Hat products used

- Red Hat OpenShift Platform Plus
- Red Hat OpenStack Platform
- Red Hat Training and Certification





How purpose-led agility is transforming Tata Tele Business Services

by Anvize Rodrigues, Chief Information Officer, Tata Tele Business Services

Tata Tele Business Services aims to be India's go-to provider of connectivity and communication solutions for businesses. We handle millions of transactions a day in the telecommunications industry, so every bit of uptime, availability, and scale is critical. However, we found ourselves held back by rigid infrastructure. We were operating as separate units, reacting to market disruptions rather than leading them.

To stay proactive and fit to face the future, we decided to completely rethink how we operated. We collaborated with Red Hat to build an open, hybrid cloud-led foundation using Red Hat OpenShift Platform Plus.

This wasn't just a simple technology purchase; Red Hat acted as a true collaborator, questioning our decisions and guiding our underlying processes and architectures to ensure that we could scale confidently.

By migrating away from traditional systems, we successfully decoupled innovation from our outdated infrastructure—and the operational and technical results speak for themselves. Provisioning times that once took 15 days have been reduced to minutes, and our billing cycles have shrunk from 10 days to just 2 or 3. More importantly, automation has drastically reduced system complexity. Instead of dealing with day-to-day operational chaos, our teams are now free to focus on strategic decision-making, innovation, and direct customer engagement.

Of course, continuous evolution in a digital-first environment can easily lead to change fatigue.

We conquer this by leading with clarity of intent—always explaining exactly why we are doing what we are doing—and implementing continuous, incremental improvements rather than disruptive, big-bang shifts. We now view organizational agility not as a reaction to a problem, but as a deliberately designed capability that is deeply embedded across our technology, culture, and decision-making processes.

Ultimately, we don't view digital transformation as a mere technology deployment. Success is measured by tangible outcomes and how well we enhance the customer journey. And by aligning our open digital platforms with a people-first mindset, our infrastructure is no longer a bottleneck. Instead, it is the dependable foundation that simplifies our customers' lives and ensures that purposeful innovation remains deeply rooted in our DNA.



TD delivers service continuity while implementing change with scalable automation

Updating disconnected systems with Red Hat Ansible Automation Platform reduces provisioning timescales from months to days without compromising stability

Delivering an uninterrupted, always-on banking experience for more than 28 million customers requires an infrastructure that is stable, consistent, and simple to update. However, TD's vast network, which covers around 2,200 branches and campuses, was growing increasingly complex due to some disconnected automation tools, inconsistent configurations, and accumulated technical debt. To standardize operations and manage its network efficiently, TD selected Red Hat Ansible Automation Platform. The new, unified approach reduced technology provisioning and upgrade timescales from months to days. By automating manual, repetitive tasks, the bank gave its network team and engineers valuable time to focus on problem-solving and innovation, improving their work-life balance while maintaining service reliability.

Modernizing network operations to support uninterrupted banking experiences

With a vast network supporting over 1,300 branches and approximately 10,000 devices, TD faced a slow and complex change-management process. Tight maintenance windows left no room for error, while manual processes forced engineers to work around the clock, impacting their work-life balance.

For a bank where even a minor misconfiguration could disrupt branch services or customer transactions, the lean team needed a way to reduce management effort without compromising trust.

The team selected Red Hat Ansible Automation Platform as the foundation for its network automation strategy due to its agentless architecture, vendor-agnostic flexibility, and ability to provide centralized visibility and control across the entire network.

Starting with a small pilot, the team methodically scaled its efforts, developing reusable playbooks, execution environments, and automated workflows to support provisioning, migrations, upgrades, and Infrastructure as Code deployments.



Increasing speed, consistency, and resilience through automation

Accelerated infrastructure updates and provisioning

With standardized, end-to-end automation, TD drastically increased speed and reliability across its infrastructure. The bank can now provision, migrate, and modify branches and campuses in days rather than months. Notably, one major infrastructure update project that previously would have taken over a year was completed in just 3.5 months. Device provisioning times plummeted from 5 hours to less than 1 hour.

Automated routine tasks for large-scale parallel changes

Routine tasks that previously required extensive engineering effort—such as port enablement, internet protocol updates, domain name system changes, documentation, and validation—are now fully automated.

This capability allows the team to execute large-scale changes in parallel across the network, delivering new capabilities faster while maintaining the strict reliability required in a high-risk banking environment.

Improved efficiency and allowed proactive development

By removing the burden of repetitive overnight changes and manual documentation, engineers can now focus on innovation and higher-value problem-solving. Automation has now become essential to the network team's daily operations. It has built more than 100 playbooks and generated 20,000 pieces of change documentation, creating a scalable framework that continuously improves with every iteration.



Expanding automation with AI and preparing for the future

To increase development speed without expanding headcount, TD's small network automation team incorporated AI copilots and custom agents to handle repetitive coding, quality assurance, and documentation tasks. The team continues to work closely with Red Hat to support continuous improvement and prepare the bank for future self-service and AI-driven capabilities.

Timeline

2022:

Initial implementation of Red Hat Ansible Automation Platform to start the network automation journey

Pilot phase:

Launched a small-scale pilot, starting with a small number of branches per night before successfully scaling up

Scaling and production:

Scaled to 50–60 locations a night, executing a major infrastructure update across over 1,300 branches and completing the project in a compressed 3.5-month period



Decreased device provisioning time from 5 hours to less than 1 hour



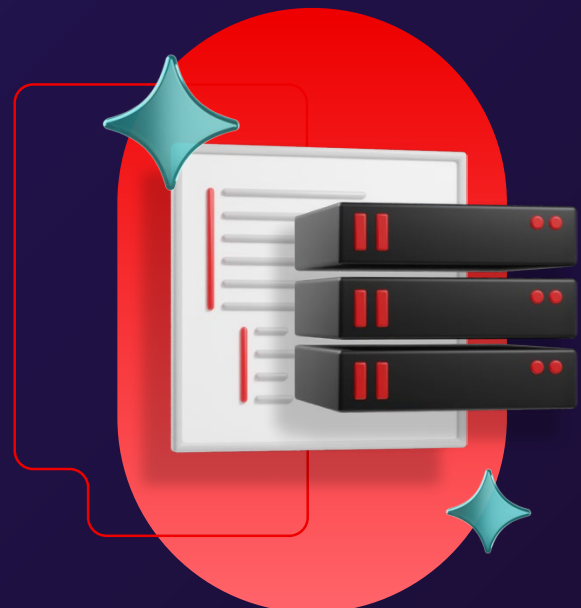
More than 100 automation playbooks were built



20,000 pieces of change documentation generated automatically

Red Hat products used

- Red Hat Ansible Automation Platform





Automation is leading TD from complex maintenance to consistent innovation

by Drew Yates, Vice President, Infrastructure, Head of Networks and Datacenter, TD

At TD, we endeavor to provide a consistent, human-centric experience for customers across North America and our global brokerages. However, supporting such a vast infrastructural footprint involves what can feel like a constant stream of lifecycle projects.

In an environment where IT infrastructure underpins every TD application and system, our engineers were burdened by manual, disconnected provisioning tasks: enabling ports, assigning internet protocol addresses, updating domain name system hosts—we did them all by hand. We also relied on a multitude of different vendor automation tools that lacked integration, forcing our lean team to execute slow, manual changes during tight maintenance windows.

To keep up with constant change, we realized that simply working even harder was no longer a viable option: we had to comprehensively rethink our approach.

We collaborated with Red Hat to decouple the complexity of our many different vendor tools and bring them into a single, cohesive platform. Our focus was to implement Infrastructure as Code using Red Hat Ansible Automation Platform, providing a vendor-agnostic view of our entire network. By building a foundation of reusable playbooks, we aimed to democratize our automation capabilities, allowing our end-user community and application owners to enact change more quickly—and without having to wait on the infrastructure team.

The results of this technical shift have been transformative. We executed a major infrastructure currency project across over 1,200 US branches in just 3.5 months—a project that typically would have taken over a year. More importantly, we significantly accelerated our speed of delivery, taking device provisioning processes that used to require 4 to 5 hours down to less than 1 hour.

By removing manual components from this workstream, we reduced the reliance on our engineering workforce for simple provisioning requests, allowing us to re-baseline our service-level agreements and shift our team's focus onto proactive platform development and higher-value innovation.

Ultimately, value in our environment is defined by making the infrastructure as invisible and unobtrusive as possible, ensuring that the bank simply works without interruption for our customers. The Red Hat Ansible Automation Platform gives us consistency and operational stability, providing a predictable platform that enriches developer agility. We are now extending these capabilities by incorporating AI assist and agentic use cases, allowing our network engineers to develop the next generation of playbooks without having to write a single line of code.

Today, our network infrastructure is a dependable partner in providing uninterrupted, high-quality banking outcomes. And thanks to Red Hat, we have shifted away from a project-only mindset to ensure that innovation is an inclusive, continuous evolution for everyone.



Conclusion

Unlocking innovation to influence an AI-led future

Whether you are carving out a competitive advantage by adopting AI, scaling into new territories, or using automation to cut costs, the potential to innovate is often tested by the friction between traditional and cutting-edge systems.

Managing hybrid environments and modern container workloads can be a heavy lift, and without a dedicated technology partner, these hurdles can easily stall the most ambitious goals. Red Hat closes the gaps, unlocking the true value of your innovation with a comprehensive portfolio specifically engineered to thrive in a complex world.

We hope that there is a customer story in this collection—such as the transformations at ARSAT, the Defense Health Agency, Emirates NBD, or StarHub—that applies to your use case or is of particular interest, and gives an insight into how others used solutions that might be relevant to your future state.

[Read](#) the success stories.